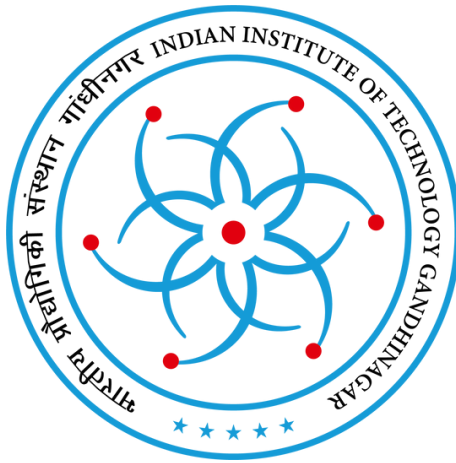




MSE203 Assignment 1

A Molecular Dynamics Study on

Melting of Gold Nanoparticles

Tensile Testing of Aluminum

Nanowire Deformation

Polymer Dynamics

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23110138

Assignment 1

I. MELTING OF GOLD NANOPARTICLES

Following is the visualization of the simulation cell in Ovito at the beginning and end of the simulation:

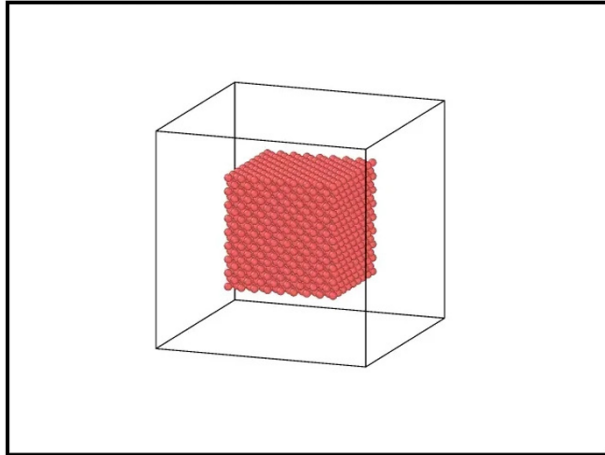


Fig. 1. Arrangement of atoms at the start of the simulation

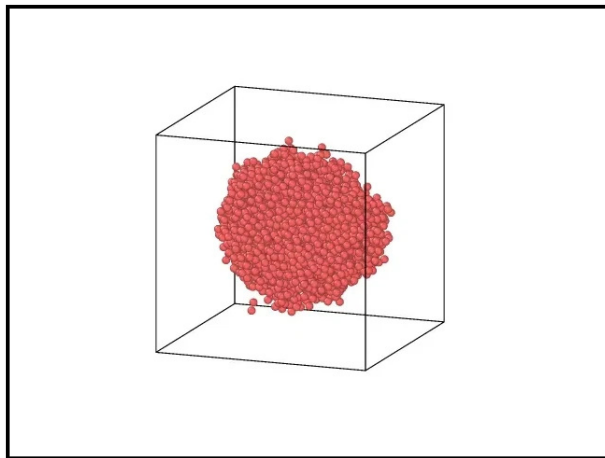


Fig. 2. Arrangement of atoms at the end of the simulation (Time-step 100/100)

Question 1(a)

Melting Point is defined as the temperature at which solid and liquid phases of a particular entity are in equilibrium^[1]. Heat supply at this temperature doesn't result in the increase in the temperature until all of the solid has converted to liquid. Hence an indication of the MP of a solid is the first appearance of disorder *which*

is sustained for some time.

Determination of this point can be accomplished by the Centrosymmetry analysis tool in Ovito.

Centro Symmetry Parameter - In solid-state systems the Centro symmetry parameter is a useful measure of the local **lattice disorder** around an atom and can be used to characterize whether or not the atom is part of a perfect lattice^[2].

The initial Centrosymmetry plot(CSP) at the start of the simulation: By monitoring the CSP plot at different

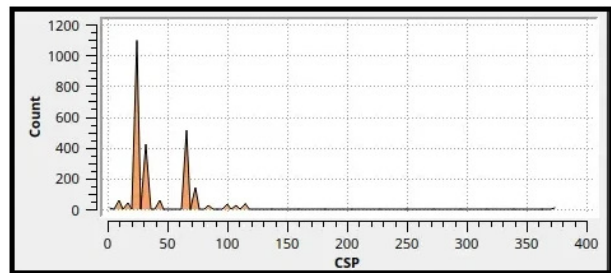


Fig. 3. Initial CSP plot indicating order

time-steps, the MP can be said to occur at a time-step such that there is a continuum in the plot, which is holds its appearance for sometime.

This is seen approximately at time-step 50/100. Analysis of the log file tells that the temperature at this time-step was 1268 K. The CSP plot at time-step 50/100 is shown below.

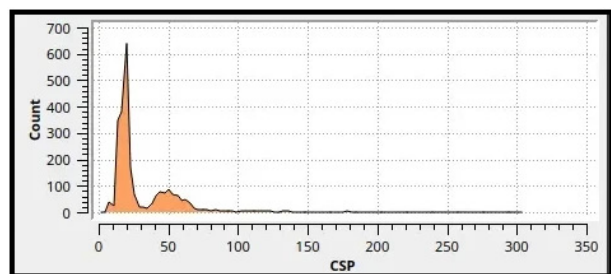


Fig. 4. CSP plot at time-step 50/100 indicating disorder

Hence, the **approximate Melting Point = 1267 K.**

The Melting point, if determined from molecular dynamics simulations, would always deviate from the thermodynamic and experimental methods. This deviation will decrease as the time-step is decreased.

Some atomic scale signatures that would validate the phase transition:

1) **Loss of Order**

Visually, there would be a loss of order. At time-step 50/100, following was the arrangement of atoms in the simulation cell:

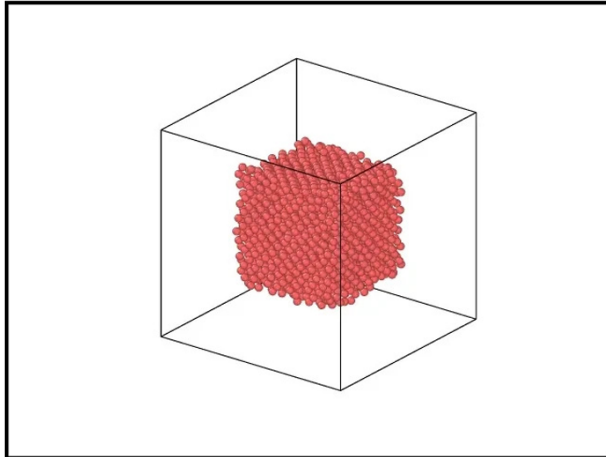


Fig. 5. Arrangement of atoms at time-step 50/100

2) **Diffusion**

Atoms would diffuse from their initial lattice positions and move about volume of the lattice.

Fig 5. clearly shows that atoms along the edges are in unconventional positions, which are not lattice points.

Question 1(b)

Figures 6 and 7 show the arrangement of atoms at two different time steps- 40 and 70/100:

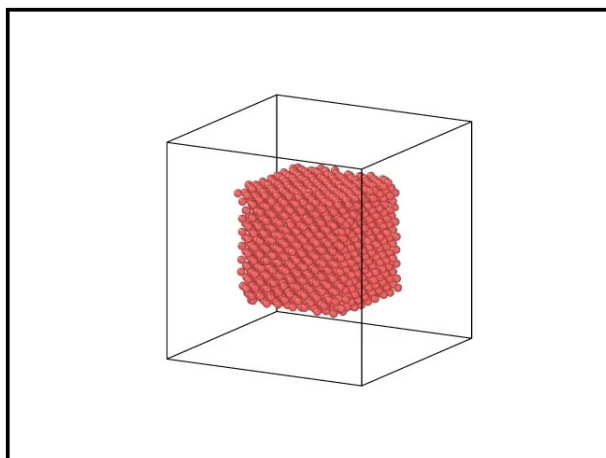


Fig. 6. Arrangement at time-step 40/100

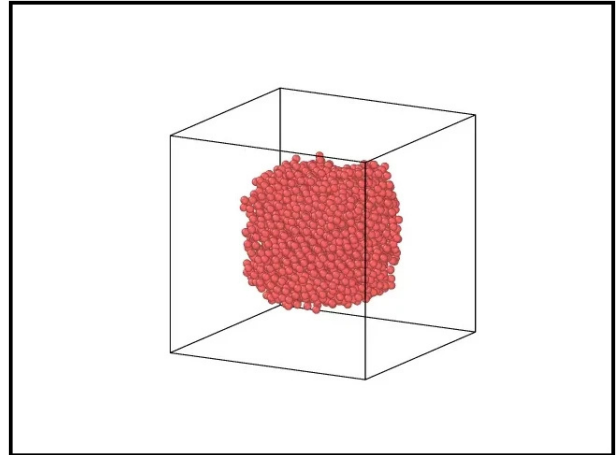


Fig. 7. Arrangement at time-step 70/100

• **Smaller Time-step (<0.001 ps)**

1. *Improved Accuracy:* Smaller time-step allows for more precise tracking of atomic positions and rapid atomic movements

2. *Better Energy Conservation:* Smaller numerical errors would result in a decrease in the numerical errors while integrating the equations of motion

3. *Phase transition:* During phase change, atoms undergo rearrangement as they diffuse around inside the simulation cell. A smaller time-step ensures these rearrangements would be captured with greater precision

• **Larger Time-step (>0.001 ps)**

1. *Reduced Accuracy:* A larger time-step would result in failure to capture fast atomic motions and hence some dynamics of the system could be missed//

2. *Energy Drift:* Larger time-step can lead to numerical instabilities causing either energy drift or poor conservation of energy over time

3. *Phase Transition:* Rearrangements of atoms would not be captured precisely

In general, the selection of smaller time-step would increase the accuracy of the results while a larger time-step would result in a decrease in the accuracy.

II. UNIAXIAL TENSILE TESTING OF MONO CRYSTALLINE ALUMINUM

Following is the visualization of the simulation cell in Ovito at the beginning and end of the simulation:

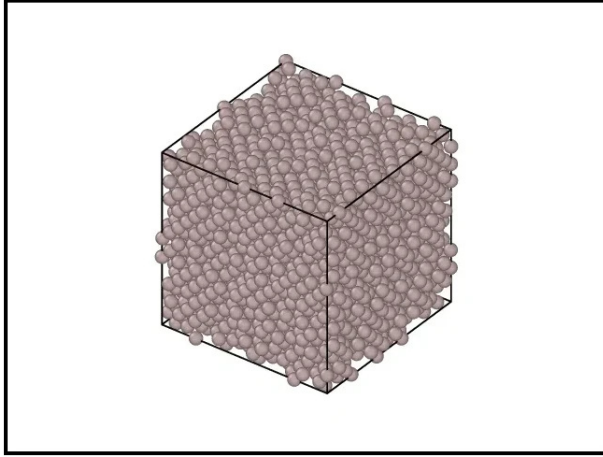


Fig. 8. Simulation cell at the beginning (Timestep 0/200)

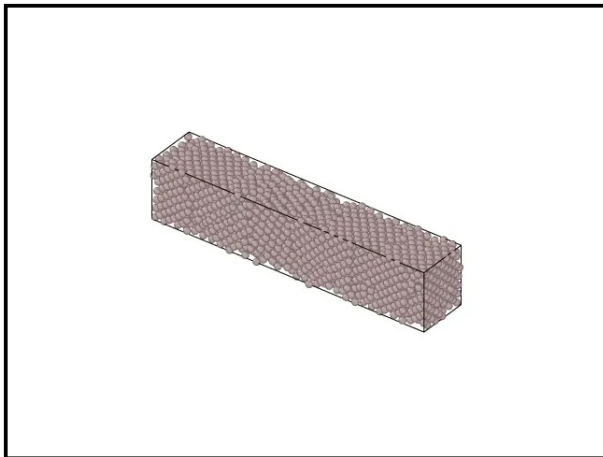


Fig. 9. Simulation cell at the end, clearly showing deformation as a result of uniaxial loading (Time-step 200/200)

Question 2(a)

The deformation was modeled at a temperature of 300 K, using the Nose-Hoover Thermostat. Figure 10 shows the resultant stress strain curve.

The elastic modulus of this material is given by the slope of the stress strain curve in the elastic region. In order to get the value of the slope a linear fit was performed on this section of the graph. The corresponding analysis is shown in Figure 11.

Approximate Yield Point = 8 bars
Elastic Modulus = 56 bars

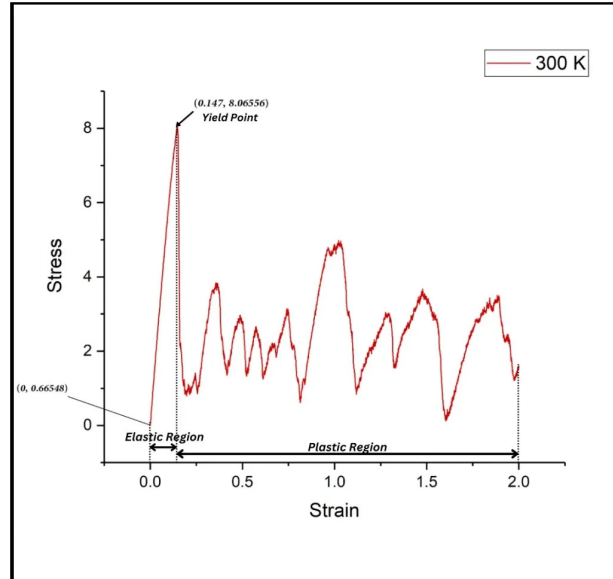


Fig. 10. Stress Strain Curve (300 K)

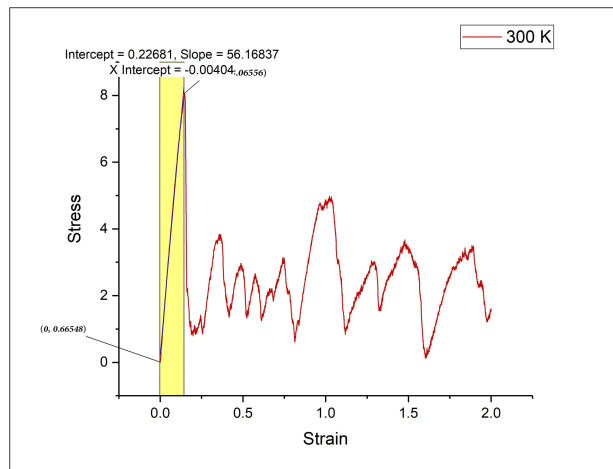


Fig. 11. Linear Fitting performed in the elastic region for the elastic modulus

Figures 12 - 17 show the state of the simulation cell at different strains. The following color code is used:

- Green - FCC
- Red - HCP
- Blue - BCC
- Yellow - Icosahedral
- White - No structure whatsoever

It can be seen clearly that the material majorly maintains its FCC structure through the course of the simulation. It is notable that other crystal structures are also formed momentarily, some of which also persist to the end of the simulation.

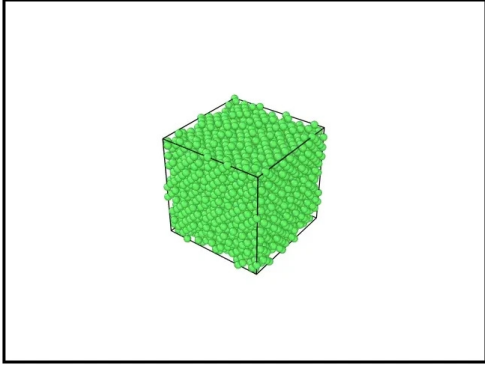


Fig. 12. Strain = 0 — FCC = 4000 — Time-step 0/200000

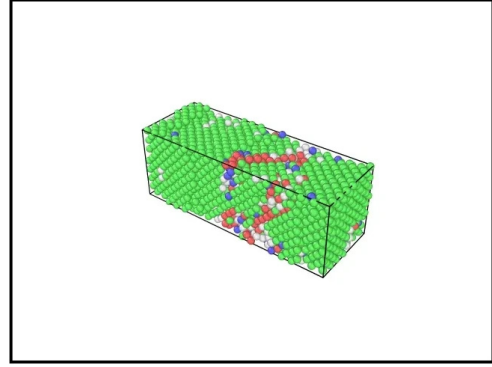


Fig. 15. Strain = 0.998 — FCC = 2598, Other = 1253, HCP = 149 — Time-step 100000/200000

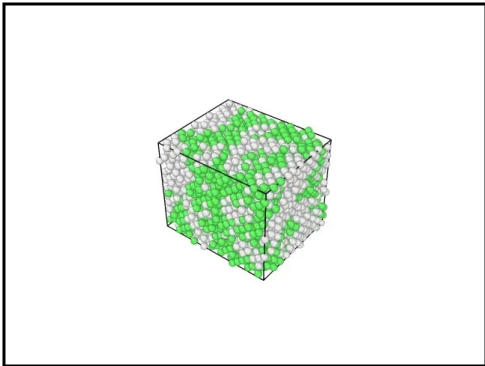


Fig. 13. Strain = 0.148 — FCC = 1869, Other = 2130, HCP = 1 — Time-step 15000/200000 (Yield Point)

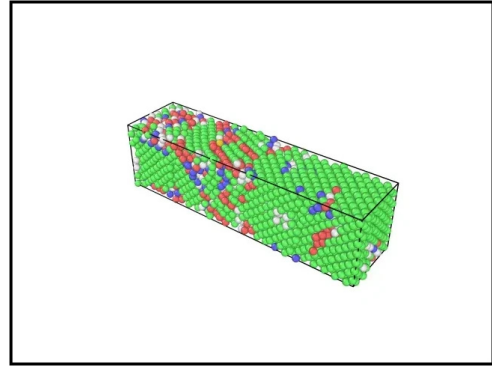


Fig. 16. Strain = 1.498 — FCC = 2005, Other = 1857, HCP = 138 — Time-step 150000/200000

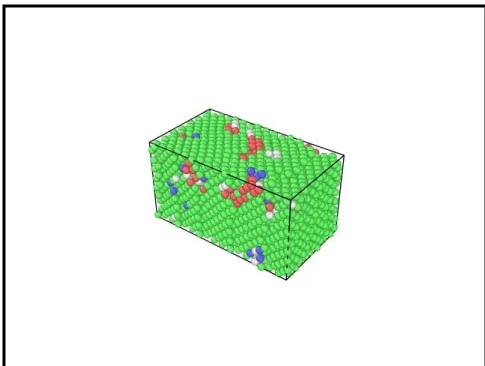


Fig. 14. Strain = 0.498 — FCC = 2771, Other = 1133, HCP = 96 — Time-step 50000/200000

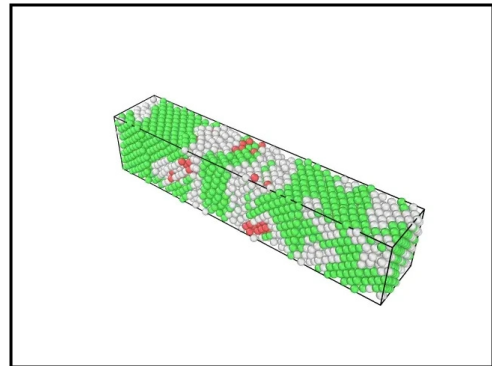


Fig. 17. Strain = 2.000 — FCC = 2244, Other = 1590, HCP = 166 — Time-step 200000/200000

Question 2(b) The stress Strain Curves at 300K, 600K and 900K are given below:

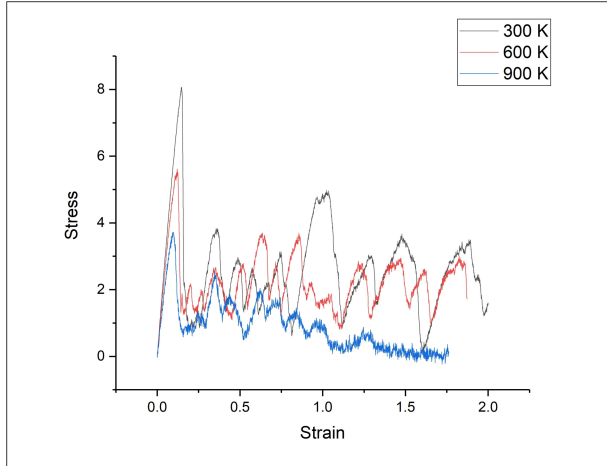


Fig. 18. Comparison of Stress-Strain Curves at 3 temperatures

In the elastic region (relevant region for each curve that is), it can be clearly seen that as the temperature increases, the slope decreases. Hence, for the same material, an **increase in the temperature results in a decrease in the modulus of elasticity**. This is due to the greater vibratory motion of the constituent atoms of the material by virtue of the higher temperature, disrupting the regular arrangement and hence making it easier for the material to deform.

It can also be inferred from the plots that, **as the temperature increases, the yield point increases**. This is because the dislocation are set into motion due to greater thermal energy.

It must also be noted that, in Fig. 10 , the region of the curve after the yield point which has been labelled as the plastic region is actually the region which occurs just before strain hardening starts. The following **complete** stress strain plot shows this region just beyond the yield point.

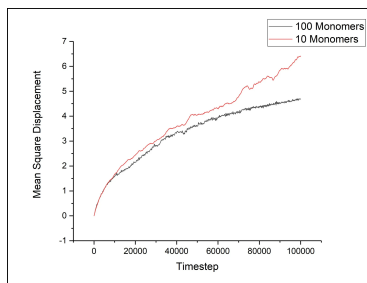


Fig. 19. A complete stress strain plot showing horizontal region just beyond yield point

III. NANOWIRE DEFORMATION

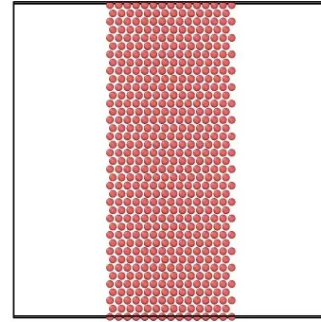


Fig. 20. Arrangement of atoms at the start of the simulation (Time-step 0/4000000)

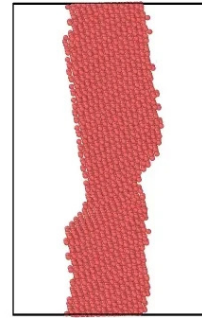


Fig. 21. Arrangement of atoms at the end of the simulation (Time-step 4000000/4000000)

It can be seen clearly from the plot in Figure 22, that deformation has indeed occurred due to the change in the z position of the atoms at the end of the simulation.

The simulation was initially scripted for 400000 runs. However, within this time frame, no noticeable changes were seen in the Nanowire model. In order to obtain viable results, the number of runs was increased to 4000000.

The stress-strain plot obtained as a result of the simulation is shown in figure 23.

Figures 24-26 at different time-steps show the formation of a neck region and figure 27 shows stacking fault.

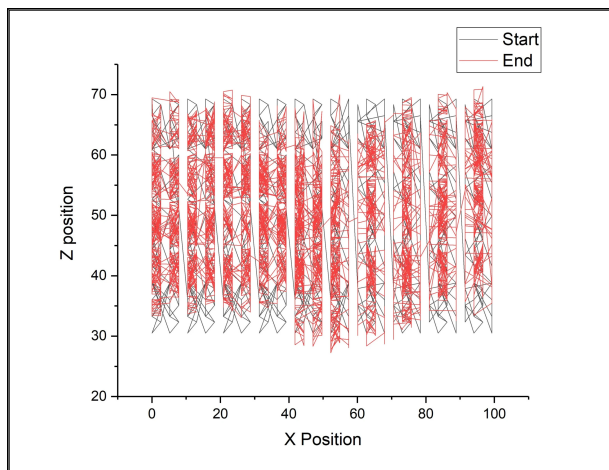


Fig. 22. Indication of deformation (analysis performed in Ovito's histogram modification)

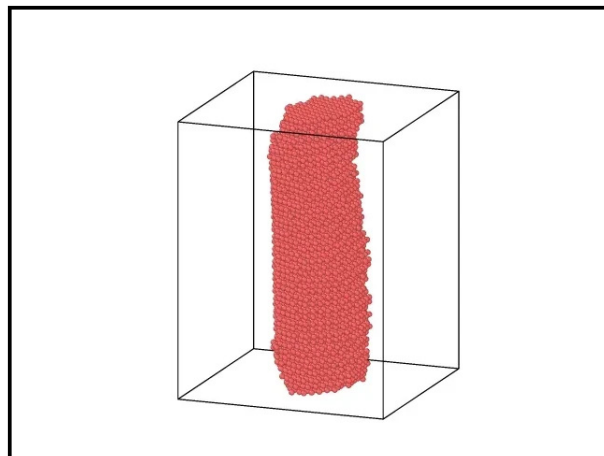


Fig. 25. Development of neck region (time-step 1602000/2000000)

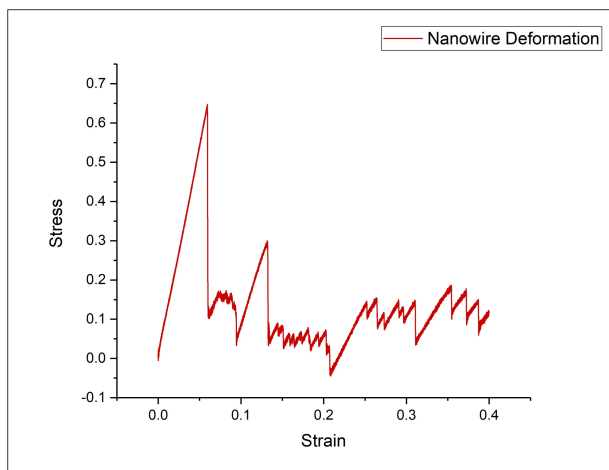


Fig. 23. Stress Strain plot for Nanowire deformation

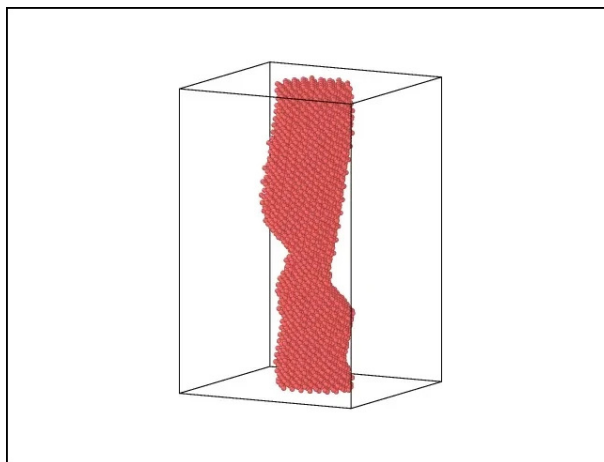


Fig. 26. Neck at the end of the simulation

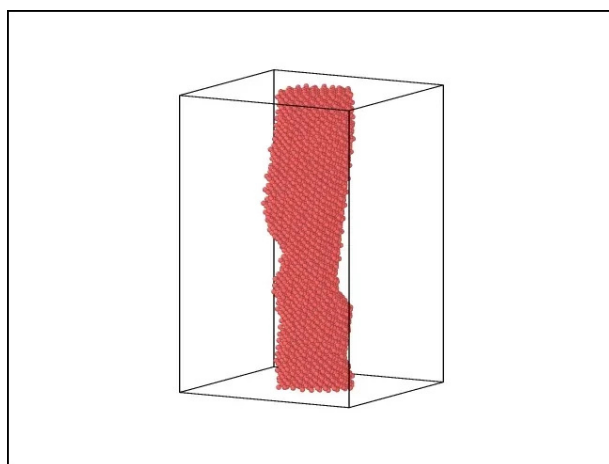


Fig. 24. Development of Neck Region

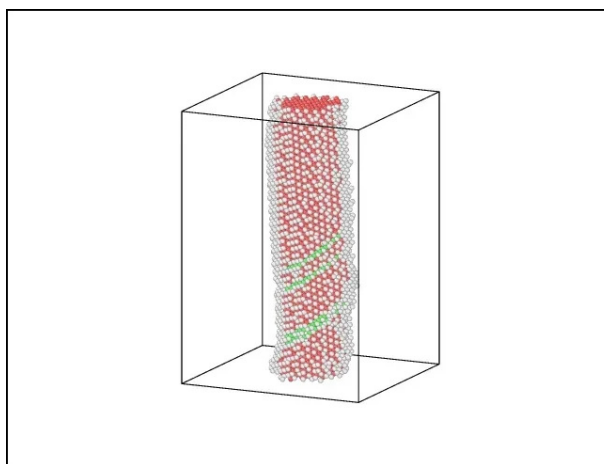


Fig. 27. Stacking fault formation; FCC - Green, HCP - Red

IV. POLYMER DYNAMICS

Question 4(a) For the polymer with 10 monomers, the MSD plot is shown in figure 27.

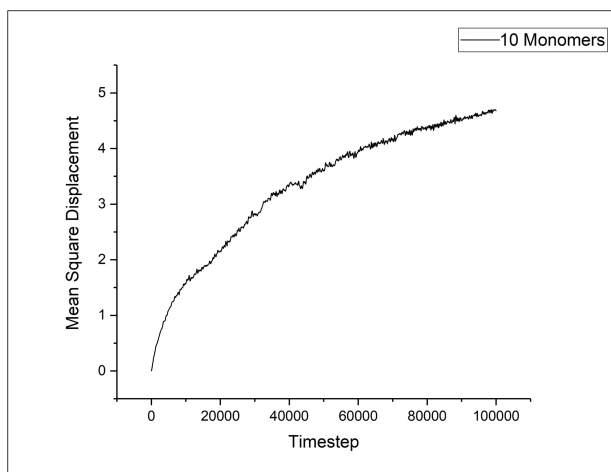


Fig. 8. MSD plot for Polymer with 10 monomers

For the polymer with 100 monomers, the MSD plot is shown in figure 28.

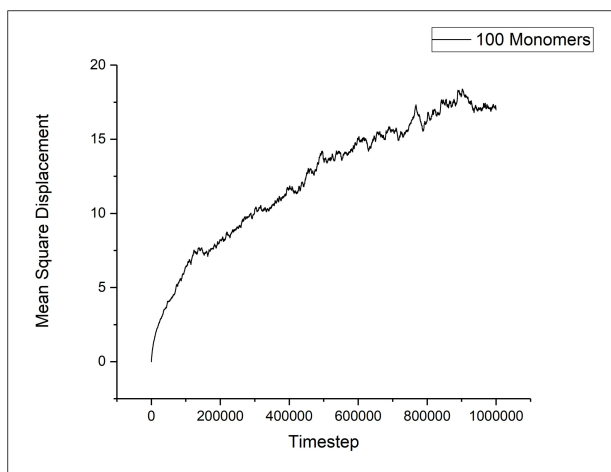


Fig. 9. MSD plot for polymer with 100 Monomers

Question 4(b) The MSD plot for the two polymers are shown together in figure 29.

It can be inferred that the polymer with 10 monomers shows a greater mean square displacement than that with 100 monomers in the same amount of time.

The polymer with 10 monomers exhibits faster molecular motion

The reason for this is the comparatively fewer 3- and 4- body interaction terms that are used to calculate the

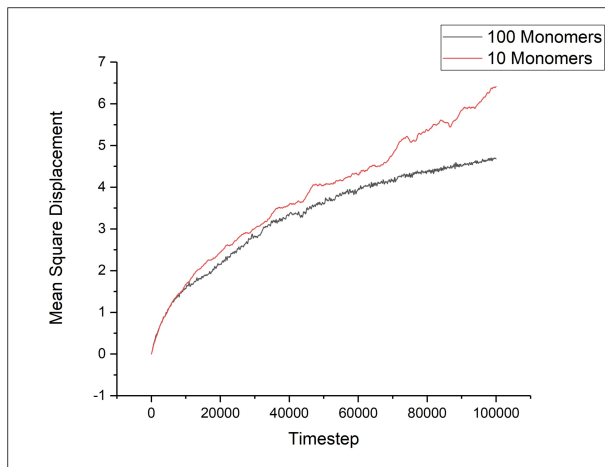


Fig. 10. Comparison of MSD plots for two polymer models

potential energy of the polymer molecule.

3 Body Terms

- In a polymer chain, these interactions contribute to the chain's flexibility and local structure
- There are fewer 3-body interactions overall
- The chain experiences less internal resistance to conformational changes
- This increased flexibility allows for greater mobility and thus a larger MSD

4 Body Terms

- These interactions are crucial for determining the polymer's overall shape and rigidity
- There are fewer 4-body interactions overall
- The reduced number of torsional constraints allows for more freedom of movement
- This increased conformational freedom contributes to a larger MSD

Another reason for this could be the drastically small entanglement that the 10-monomer polymer chain experiences whilst moving, as compared to the 100 monomer chain.

V. RESOURCES AND REFERENCES

A. Resources

All of the **input scripts** pertaining to each question in this assignment can be found on the [following](#) GitHub Repository, in addition to the results of the simulation.

Normally, simulating the systems in the assignment, notably that pertaining to question 3, was taking a long time. In the pursuit of saving this time, the simulations were all performed using **320 cores of T4 GPU** available for use through Google Colab. The code for the same can be found [here](#).

The help of my friend Nilay Verma (23110219) and Rudra Pratap Singh (23110281) were invaluable in this approach.

Origin has been used to make all the plots in this report. The associated project files for Questions 2,3, and 4 can also be found in the GitHub repository.

B. References

- 1) [Lammps Documentation](#)
- 2) [Ovito Documentation](#)